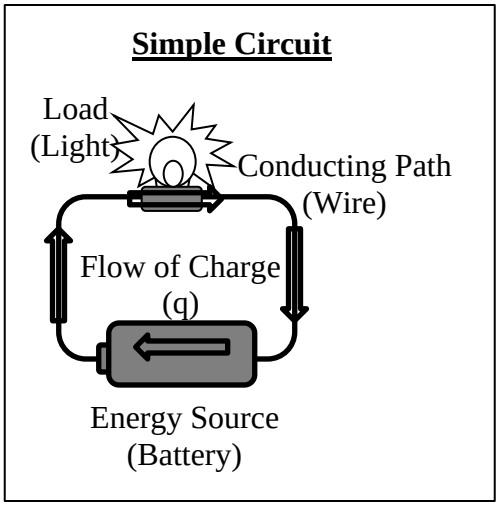
Chapter 41 – Current, Resistance & Ohm's Law
Calculus Physics II

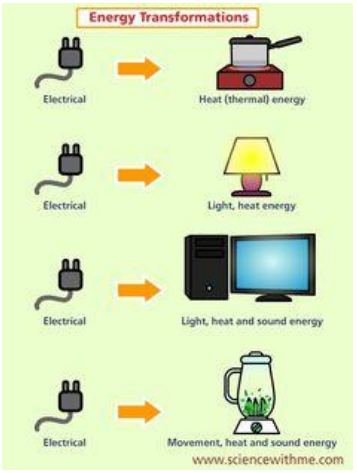
Using the Energy of Charged Particles - By doing work on charged particles, these charged particles can be separated, this charge separation creates an electric field; when charged particles are placed in this electric field, they experience a force and have a potential energy, this energy can be used to do work. As the charge moves along the electric field this Electrical energy can be converted to other forms of energy (mechanical, thermal, light, etc. …), this chapter looks at how this energy is used by the electric circuit.



Circuit – any conducting path that charge can flow. More specifically, it is the total electrical path made of conducting wire, power sources and circuit elements.

Load – is what is being run by the power source in the circuit. The load uses the electrical power that the source supplies.

Electrical Power source – any device that separates (or collects) charge by use of mechanical or chemical or other methods to create the electric field. The power source does work to give the charge a potential energy (PE_{elec}) and a potential (V) due to the electric field. Examples: Batteries, electric generators, solar cells, capacitors, etc. …



- The amount of energy available from a power source that supplies a constant potential (voltage) is the amount of energy each until charge has, which is the same for all charge because all the charge is at the same potential therefore at the same potential energy. To get more energy out of a power source the amount of charge supplied by the power source must increase.

- A power supply that the potential (voltage) can be changed, when the voltage (potential) is increased the amount of potential energy of each of the charges increases, thereby increasing the amount of energy available to the circuit to increase the electric field strength.

Electromotive Force (emf) – The ideal electric potential difference that exists between the terminals of an electrical power source. The emf is the ideal potential, ignoring any loss of energy. The emf is the work done by the electric force on each charge in the electric field.

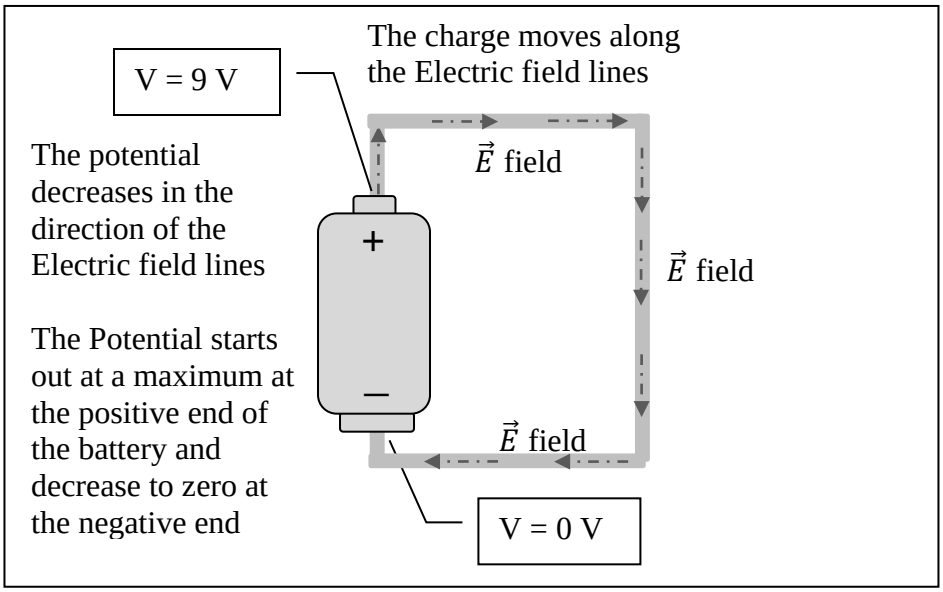
- Units are Volts (V)
- The emf is equal to the terminal voltage for a perfect power source.
- The emf is not a force but a potential energy per unit charge, it is the amount of potential energy available to move the charge (unit charge) in the circuit.

- The emf is the energy that the charge has because of the work the power source does on the charge, it is an electric potential.

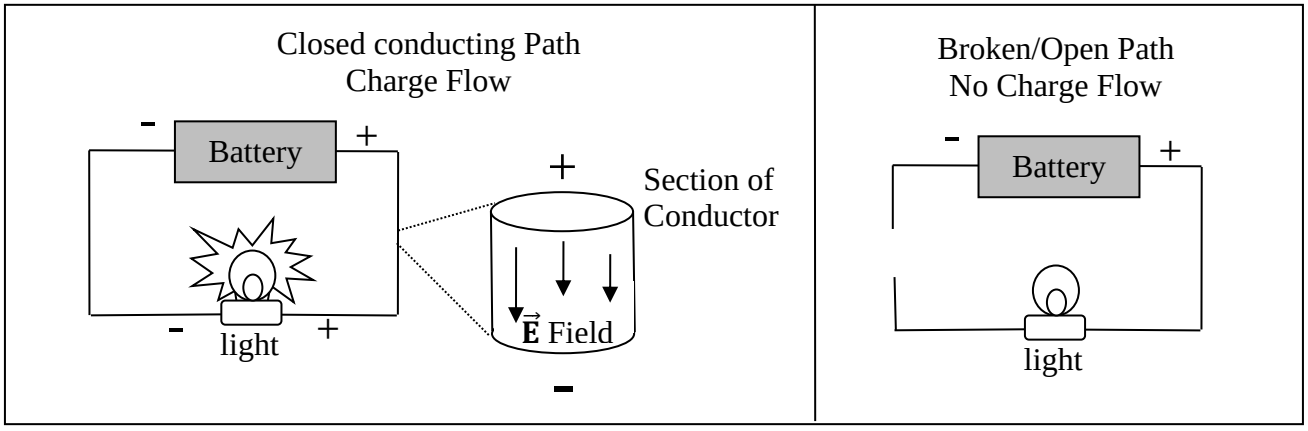
$$emf = \frac{Energy}{charge} = \frac{\Delta PE_E}{q} = V_{Ideal}$$

What moves the charge in a wire? If there is no Electric Field in the conducting wire, then the electrons move randomly. When there is an Electric field in a conductor (wire) the $\vec{E}$-field provides the force that moves the electrons (charge) through the circuit.

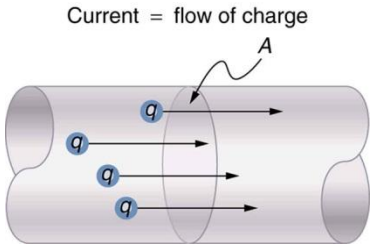
The electric field is established at the speed of light (3 x 10^8 m/s) in the wire so as soon as the conducting path is switched "on" the electric field is nearly instantaneously set up in the wire driving the charge in all parts of the circuit almost instantaneously. It is the Electric field that is doing the work in moving the charge in the circuit.



- The electric field creates the potential (and potential energy) in the circuit to do the work that runs the electronic devices connected to the power source.



Current (I) – a flow or movement of charge - An electrical current exists in a region when a net electric charge is transported from one point to another by the force due to the electric field. Current is a scalar quantity.



$$I = \frac{dq}{dt}$$

Units are Ampere or Amps (A) 1A = 1 C/s

Where q is the charge and t is the time in seconds.

- Current is the amount of charge per unit time through a cross-sectional area.
- When the charge flows out of one section it is replaced by that same amount of charge from the other end.
- In order to have a steady flow of charge there must be an electric field present.
- Current is always through a circuit and Potential (voltage) is always across.
- Current can be any charged particle, electrons, protons or atoms that are ionized, it does not have to be only electrons, but electrons are the most common charged particles to produce current.

➢ For charge to continuously flow in a conductor it must be replaced by other charge, if not it will stop flowing.

Example: How many electrons are moving through the cross-sectional area of a wire with 1 amp of current in 60 seconds?

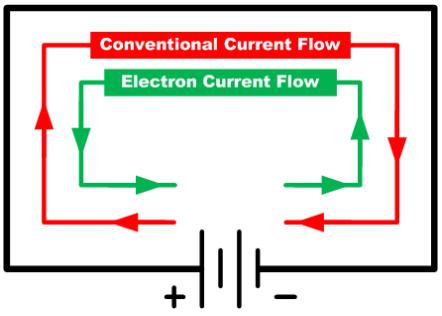
$I = \Delta q/\Delta t$ therefore $\Delta q = I\Delta t = (1\ A)(60s) = 60\ C$

$N_e = Q/q_e = (60\ C)/(1.6 \times 10^{-19}\ C/e^-) = 3.75 \times 10^{20}$ electrons

Direction of Current Flow - By convention (not actual) the direction of the current flow is taken to be the direction of flow of positive charge (from positive to negative). In a real circuit it is the electrons not protons or positive charge that does the moving (electrons flow from negative to positive).

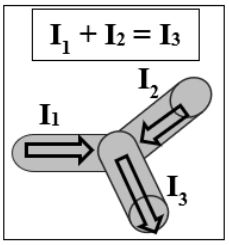
Conventional Current – The hypothetical flow of **positive charges** that would accomplish the same effect in the circuit as the movement of negative electrons that actually does occur in the wire.

Conventional current (positive charge) flows from positive to negative.



Conservation of charge – Electric charges can be produced only in pairs of equally strong positive and negative charges, therefore the total charge in a closed system does not change. Charge is neither created nor destroyed it is transferred from one object to another.

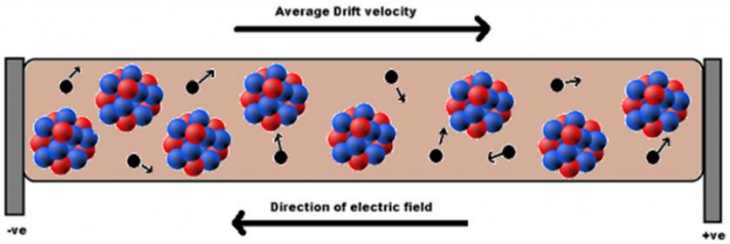
Conservation of Current – The total current in a closed circuit does not change. The total current going into a closed circuit must be the same as the total current out of that circuit.



- The current can divide at junctions and take different paths in the circuit, but the total current does not change.
- This is a direct result of the conservation of charge.

Drift velocity (v_d)– The actual speed of the electrons in the wire is very slow (10^{-4} m/s) because they are constantly running into atoms in the wire. The average speed of the electrons in the circuit is called the drift speed.

It takes almost 3 hours for a single electron to go through 1 meter of wire but when you turn a light on it appears instantaneous, how? The electric field in the wire propagates at near the speed of light (3 x 10^8 m/s) and it is the field that produces the force on the electrons throughout the entire wire. Every charge in the wire "feels" the electric force due to the electric field.



Current Density - Current density is a vector quantity that describes the flow of electric charge in a given area of a conductor. It is denoted by the symbol $\vec{J}$ and is defined as the amount of electric current flowing per unit area perpendicular to the flow.

$$I = \iint \vec{J} \bullet d\vec{A} \qquad |\vec{J}| = \frac{I}{A}$$

$\vec{J}$ is the current density (in amperes per square meter, A/m²),
I is the electric current (in amperes),
A is the cross-sectional area through which the current flows (in square meters).

Current density is a vector, meaning it has both magnitude and direction. Its direction is the same as the direction of the flow of positive charges (conventional current). On a microscopic level, current density is related to the drift velocity of charge carriers:

$$\vec{J} = nq\vec{v}_d$$

n is the number of charge carriers per unit volume (units $1/m^3$)
q is the charge of each carrier (C)
$\vec{v}_d$ is the drift velocity of the carriers (m/s)

This equation shows how current density depends on the material properties and the behavior of the charge carriers inside the conductor.

Resistance (R) – Impedes the flow of charge/current**.** The Resistance of a material can be considered nearly constant in most cases and is a measure of the potential difference ΔV that must be applied across the object to cause a current of one ampere to flow through it.

- Units: Ohms (Ω)
- Resistance is a property of the circuit (wire and elements in the circuit)

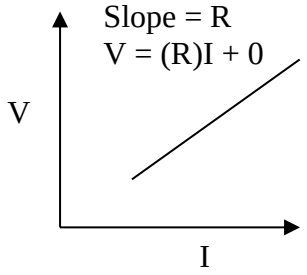
Conductivity (G) – How easily charge flows. G = 1/R Units; mhos or siemens

Resistor – an electrical device that produces a resistance to the flow of charge.

Ohm's Law - Where ΔV (or just V) is the potential difference V between the two ends of a resistance R, I is the current through the resistor and R is the resistance under those conditions.

$$\Delta V = IR \quad \text{or} \quad V = IR$$

- If voltage (V) versus current (I) are plotted for some circuit element, and if the line is straight, then the slope is the resistance, and it has a constant slope which means a constant resistance.
- Ohm's Law is actually more of a rule of thumb.
- The resistance in a circuit **must be constant to apply Ohms Law**.



Example: A car battery creates a 12 V drop across the headlights that have a resistance of 8 ohms, what current flows in order to run the lights?

$$V = IR \quad \text{therefore } I = V/R = 12\ V/8\ \Omega = 1.5\ A$$

Resistivity (ρ) - is the property of the material that impedes the flow of charge and is different for different materials. The resistance (R) in ohms of a wire or other material of length L in meters and cross-sectional area A in meters squared, the resistance of a wire is given as

$$R = \frac{\rho L}{A}$$

- Where ρ (rho) is the resistivity and is a constant for a given material.
- Units of Resistivity are Ohm meters (Ωm)

- ✓ Electrical Insulating materials have high resistivity.
- ✓ Electrical Conductors have low resistivity.
- ✓ An Ideal conductor has zero resistance.
- ✓ An Ideal insulator has infinite resistance.

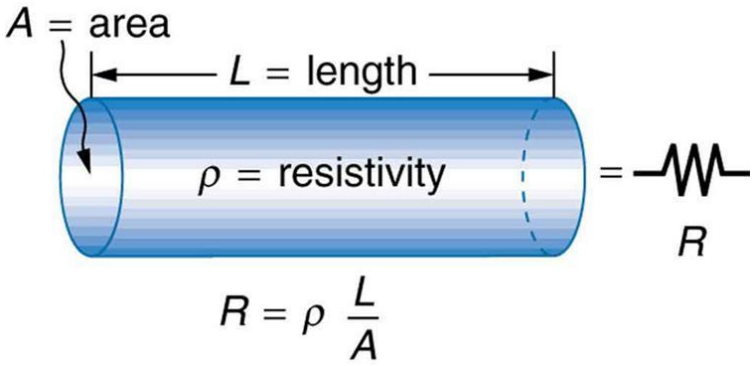


Table of some common material and their Resistivities.

Material	Resistivity ρ ($\Omega \cdot m$)	Material	Resistivity ρ ($\Omega \cdot m$)
Conductors		***Semiconductors***	
Aluminum	2.82×10^{-8}	Carbon	3.5×10^{-5}
Copper	1.72×10^{-8}	Germanium	0.5[b]
Gold	2.44×10^{-8}	Silicon	20–2300[b]
Iron	9.7×10^{-8}	***Insulators***	
Mercury	95.8×10^{-8}	Mica	10^{11}–10^{15}
Nichrome (alloy)	100×10^{-8}	Rubber (hard)	10^{13}–10^{16}
Silver	1.59×10^{-8}	Teflon	10^{16}
Tungsten	5.6×10^{-8}	Wood (maple)	3×10^{10}

Example: How much current would flow through a 100 meter copper wire with a cross sectional area of A = 3.3×10^{-6} m^2 if a 9 V battery was attached across the wire?
$\rho_{copper} = 1.72 \times 10^{-8}$ Ωm

$R = \rho L/A = (1.72 \times 10^{-8}\ \Omega m)(100\ m)/(3.3 \times 10^{-6}\ m^2) = 0.5\ \Omega$

V = IR solving for current $I = V/R = 9\ V/0.5\Omega = 18\ A$

Current Density and Ohm's Law - Current density also provides a more fundamental way to express Ohm's Law. For many materials, the current density is directly proportional to the electric field $\vec{E}$ that drives the charge flow. The constant of proportionality is the material's resistivity (ρ). This equation is known as the microscopic form of Ohm's Law.

$$\vec{J} = \rho\vec{E}$$

Temperature dependence of Resistivity – for most materials as they are heated (adding thermal energy) the resistivity of the material goes up and conversely as they are cooled their resistivity goes down. This change in resistivity depends on the change in temperature and is given by

$$\rho = \rho_o [1 + \alpha(T - T_o)]$$

where α is the temperature coefficient of resistivity, when resistivity increases with increasing temperature then α is positive.

Temperature dependence of Resistance $R = R_o [1 + \alpha(T - T_o)]$

material	α - Coefficient of resistivity (/°C)
Tungsten	$+ 4.5 \times 10^{-3}$
Copper	$+ 4.0 \times 10^{-3}$
Iron	$+ 5 \times 10^{-3}$
Ni-chrome	$+ 0.4 \times 10^{-3}$
Carbon	$- 0.5 \times 10^{-3}$
Silicon	$- 75 \times 10^{-3}$

- Note that Carbon and Silicon have negative coefficients, which means their resistivity goes down with an increase in temperature.

Example: Ni-chrome wire is used in heating elements, what is the difference in resistance of a Ni-chrome wire ($A = 2 \times 10^{-7}$ m^2), length of 10 meters at room temperature (20 °C) and when the wire is heated to 1,000° C? $\alpha = 0.4 \times 10^{-3}$/°C

The resistance R_o of the wire at room temperature

$$R_o = \rho L/A = (100 \times 10^{-8}\ \Omega m)(10\ m)/(2 \times 10^{-7}\ m^2) = 50\ \Omega$$

The resistance R_{hot} of the wire at 1,000 °C

$$R_{hot} = R_o [1 + \alpha(T - T_o)] = 50\ \Omega[1 + 0.4 \times 10^{-3}/°C(1{,}000\ °C - 20\ °C)$$

$$R_{hot} = 69.6\ \Omega$$

The difference if the resistance from hot to room temperature

$$\Delta R = R_{hot} - R_o = 19.6\ \Omega \text{ increase with temperature}$$

What would happen to the current in the wires as the wires heat up if the voltage drop across the wires is 120 V?

$$I_{cool} = V/R_o = 120\ V/50\ \Omega = 2.4\ A$$

$$I_{hot} = V/R = 120\ V/69.6\ \Omega = 1.7\ A$$

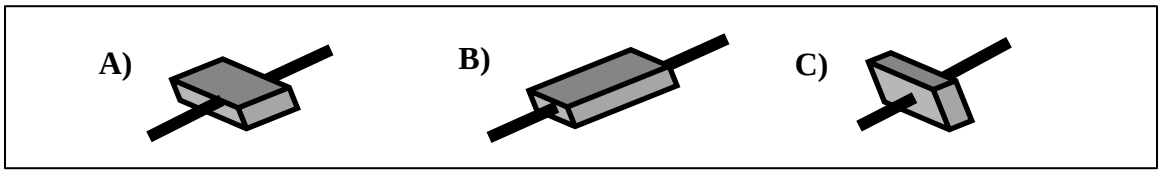
The wire heated up the resistance of the wire increases and the current decreases.

Difference between Resistance and Resistivity

- Resistivity depends on the material the device is made of (copper, steel, aluminum, etc …)

- The Resistance depends on the geometry and shape of the device (length, cross section, material)

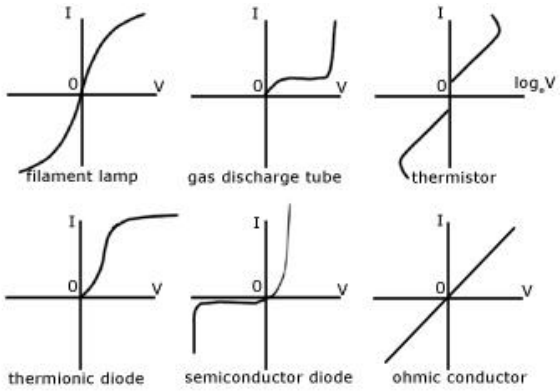
Example: For a copper block that is placed in a circuit three different ways, rank from largest to smallest the result of the resistance and resistivity to the circuit.



- Resistance is Largest in B) because of the smaller cross-sectional area and the longer length and smallest is C) because of the larger cross-sectional area and the shorter length.
- Resistivity is the same for all because Resistivity only depends on the material not the geometry.

Non-Ohmic Materials – The materials do not obey Ohms Law, the slope of I vs. V is not constant.

Examples: Semiconductors, diodes, light bulb when it first starts ...



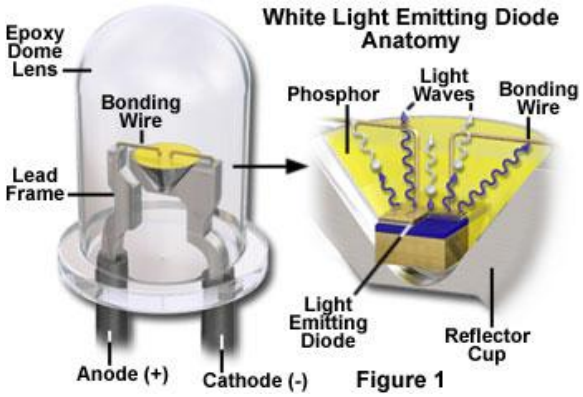
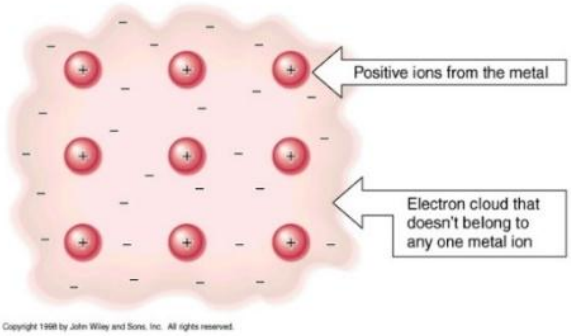


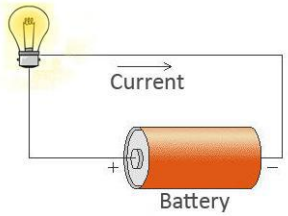
Figure 1

Metals and conductivity – Metals make good conductors of charge because of the outer most electron in the metal are not tightly bound to the atom and are free to move known as the "sea of electrons". Therefore, any electric field will create a movement of electrons (current).



Simple Circuit - a simple circuit consists of a power sources (electric potential) and a component (load) that uses the power to perform some function.

Example #1: An electric field is established through the circuit (wire and light bulb) by the battery. The same amount of current (number of charged particles) is coming out of the positive end of the battery that goes back into the battery at the negative end. The charge moving through the filament gives up their energy as thermal energy that heats up the filament.



- The charge (the electrons) is NOT used up, the energy that the charge has is what is used and gives the light its energy!

Ideal wire – There is no loss of energy as current flows through an ideal wire therefore there is no potential drop along or across an ideal wire.

Branches in a simple circuit - For a simple circuit with branches (nodes) the total current going into the branch is split into two different currents (point A in the diagram) and the amount of current in each branch depends on the resistance in that branch of the circuit. But the total current going into the branch is equal to the sum of the currents coming out.

$$I_{in} = I_1 + I_2 = I_{out}$$

For point B in the diagram the current recombines to give the same current as the original current leaving the power supply.

